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Hallucination Neurons and Where to Find Them

An Investigation into the Existence of Hallucination Neurons in LLMs via Diagnostic Protocol

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A five-step diagnostic protocol reveals that 19 of 22 claimed hallucination neurons in Gemma 3 4B have Pearson $|r| > 0.7$ with other features, exhibit only moderate bootstrap stability, and show weak sparse-dense ranking overlap — demonstrating that sparse predictive structure coexists with non-unique neuron selection, and that routine diagnostic validation is necessary before making causal localization claims.

AT A GLANCE

Problem: Sparse probing claims specific neurons causally drive LLM hallucination, but these claims are not tested against known failure modes of L1-regularized probing in correlated feature spaces.

Why it matters: Non-unique neuron selection undermines the validity of targeted steering and auditing based on claimed hallucination neurons.

Method: A five-step diagnostic protocol: feature correlation, bootstrap stability, sparse vs. dense ranking, intervention baselines, and cross-dataset evaluation.

Result: 19/22 H-Neurons fail the feature correlation criterion in Gemma 3 4B; bootstrap stability and sparse-dense agreement are weak.

Evidence: TriviaQA, BioASQ, and NQ-Open across Gemma 3 4B and MedGemma 4B.

The Big Picture

Mechanistic interpretability of large language models has made rapid progress in identifying circuits, features, and neurons associated with specific behaviors. A particularly consequential line of work claims to identify “hallucination neurons”—small sets of neurons whose activation levels predict and causally drive factual hallucinations. If such neurons exist and are uniquely identifiable, they offer a compelling target for behavioral steering: suppress the neuron, suppress the hallucination.

But neuron identification typically relies on L1-regularized (Lasso-style) sparse probing, which is well-known to select non-unique solutions in the presence of correlated features. A Lasso probe that achieves high AUROC does not estab-

lish that the selected neurons are the unique causal locus of the behavior—only that they are predictive given the regularization and the training data. In high-dimensional, highly correlated neural activation spaces, this distinction matters enormously.

This paper proposes a minimum diagnostic standard for neuron-localization claims and applies it to the H-Neuron literature, revealing systematic non-uniqueness in the claimed hallucination neurons.

Why Existing Approaches Fall Short

L1-regularized probing selects solutions non-uniquely when features are correlated: many alternative neuron sets achieve similar or better discrimination. The selected set depends on regularization strength, initialization, and the specific training instances.

High AUROC is a necessary but not sufficient condition for localization. Many overlapping neuron subsets may achieve high AUROC; the probe’s predictive success does not establish which neurons are causally necessary.

Ablation studies (removing a selected neuron and measuring behavior change) are often compared against no-ablation baselines rather than against equally predictive alternative neurons, making causal claims fragile.

Single-dataset evaluation cannot distinguish neurons that detect hallucination generically from neurons whose predictive power is specific to the evaluation dataset’s format or domain.

Core Mechanism

Five-Step Diagnostic Protocol. The paper proposes five diagnostic criteria that should be satisfied before claiming unique neuron localization:

1. **Feature correlation:** Flag selected neurons with Pearson $|r| > 0.7$ with unselected features as potentially non-unique.
2. **Bootstrap stability:** Measure Jaccard overlap of the selected neuron set across bootstrap samples of the probe training data.
3. **Sparse vs. dense ranking:** Compare the ranking of neurons by the L1 probe with their ranking by a dense (non-regularized) importance score; low overlap signals L1-driven selection, not importance.
4. **Intervention baselines:** Compare ablation of selected neurons against ablation of equally predictive randomly drawn neurons; a uniquely causal neuron should show strictly greater behavior change.
5. **Cross-dataset evaluation:** Evaluate selected neurons on held-out datasets not used for selection; poor transfer indicates dataset-specific rather than general hallucination neurons.

Technical Formulation

Let $\mathbf{h} \in \mathbb{R}^d$ be activations at a layer, $y \in \{0, 1\}$ the hallucination label. The L1-regularized probe solves:

$$\hat{\beta} = \arg \min_{\beta} \mathcal{L}(y, \mathbf{h}^\top \beta) + \lambda \|\beta\|_1$$

Selected neuron set: $S = \{i : \hat{\beta}_i \neq 0\}$.

Criterion 1 (feature correlation): Flag S if $\exists j \notin S, i \in S$ such that $|\text{Corr}(h_i, h_j)| > 0.7$.

Criterion 2 (bootstrap stability): $J_B = |S \cap S_b| / |S \cup S_b|$ for bootstrap sample b ; flag if $\mathbb{E}[J_B] < 0.8$.

Criterion 3 (ranking disagreement): Spearman $\rho(\hat{\beta}, \tilde{\beta})$ where $\tilde{\beta}$ is the dense-probe importance; flag if $\rho < 0.5$.

Cross-dataset AUROC gaps (Gemma 3 4B vs MedGemma 4B):

$$\Delta_{\text{TriviaQA}} = +0.311 \text{ vs } +0.235$$

$$\Delta_{\text{BioASQ}} = +0.474 \text{ vs } +0.455$$

$$\Delta_{\text{NQ-Open}} = +0.128 \text{ vs } +0.112$$

Learning or Inference Procedure

The diagnostic is applied post-hoc to trained probes:

1. Train an L1 probe on each (model, dataset) pair.
2. Record the selected neuron set S .
3. Apply each diagnostic criterion.
4. Report pass/fail and overall diagnostic verdict.

No model training is performed; the protocol is a validation framework.

What the Guarantee Says

When all five criteria are satisfied, the selected neuron set is consistent with being a unique causal locus of the behavior. When criteria 1–3 fail (as they do here), the probe’s predictive power is consistent with a broader class of alternative neuron sets, and the causal localization claim is not established. This does not falsify hallucination detection; it shows the detection is distributed, not localized.

Experimental Findings

- **19 of 22** H-Neurons in Gemma 3 4B have Pearson $|r| > 0.7$ with at least one unselected feature—failing criterion 1
- Bootstrap Jaccard overlap is moderate, confirming sensitivity to training subsample—failing criterion 2
- Sparse-dense rank correlation is weak, indicating L1 regularization drives the selection—failing criterion 3

- Gemma 3 4B consistently outperforms MedGemma 4B on hallucination probe AUROC across all three datasets
- Cross-dataset probe quality varies, suggesting partial dataset specificity in neuron selection

Ablations and Interpretation

AUROC vs. uniqueness: High AUROC is achieved by many overlapping neuron sets; the reported probe AUROC does not support localization.

Dataset specificity: Different datasets select partially different neuron sets, consistent with distributed rather than universal hallucination neurons.

Implications for steering: Interventions on L1-selected neurons may or may not generalize because many correlated alternatives exist. Robust steering likely requires denser probes, ensemble methods, or causal identification beyond predictive accuracy.

Broader message: Routine diagnostic validation—particularly feature correlation and bootstrap stability checks—should be a minimum requirement for sparse-neuron localization claims in interpretability research.

Reference

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